# Marshrutlash protokollariga umumiy nazar

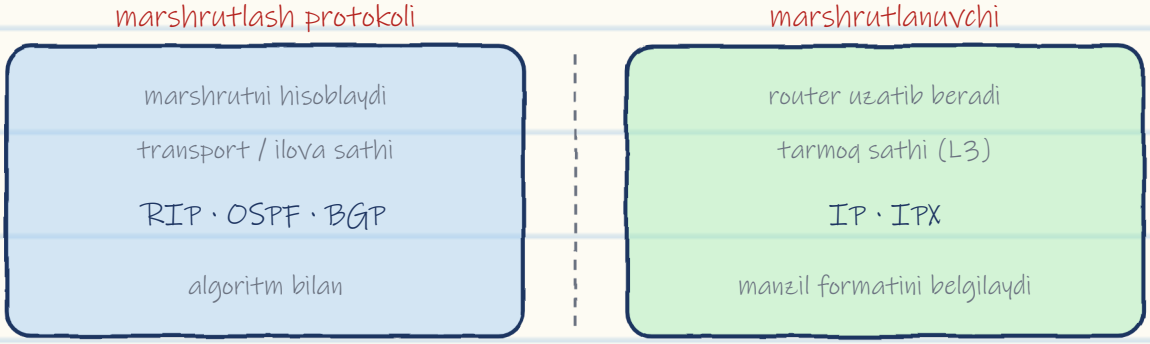
## Overview of Routing Protocols

Asosiy atamalar

- Marshrutlanuvchi protokol — uzatiladigan ma'lumot protokoli, masalan IP
- Marshrutlash protokoli — marshrutlarni topib beradi: RIP, OSPF, BGP
- AS (avtonom tizim) — bitta boshqaruvdagi tarmoqlar guruhi
- IGP / EGP — AS ichidagi / AS'lar orasidagi protokol
- Masofa-vektor — qo'shni aytgan masofaga ishonadi (RIP)
- Kanal holati — butun topologiyani biladi, SPF hisoblaydi (OSPF)

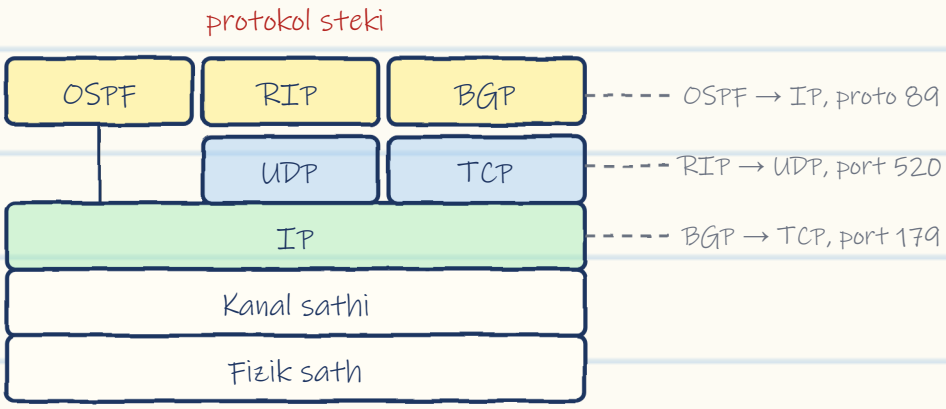
## 1. Ikki xil "protokol"

Marshrutni protokol hisoblaydi, paketni esa boshqa protokol tashiydi.




1-rasm. Biri yo'lni topadi, ikkinchisi yo'ldan yuradi

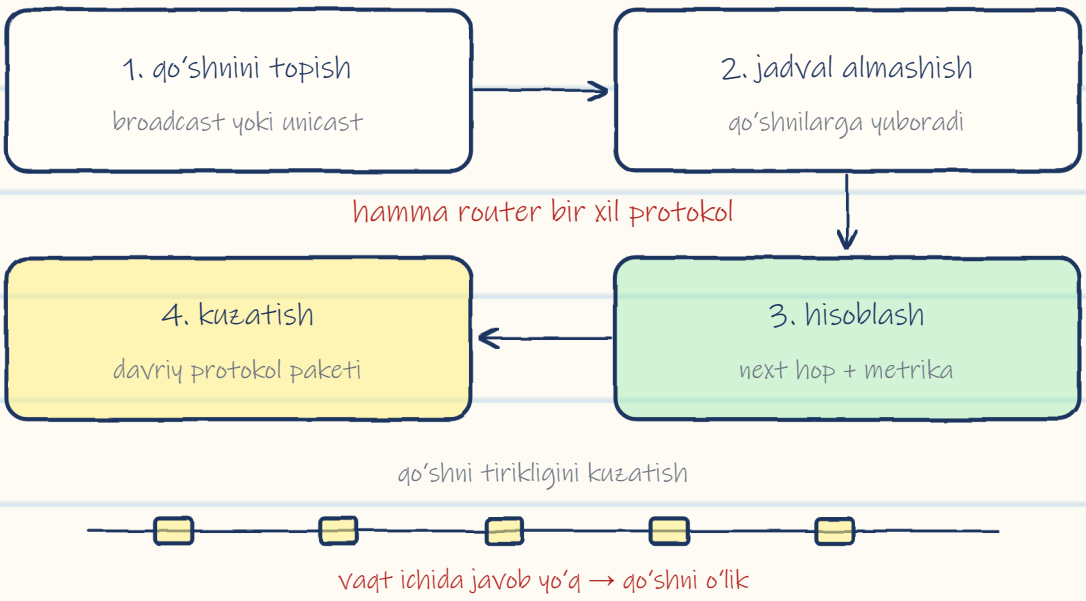
## 2. Protokol stekidagi joyi




2-rasm. OSPF ishonchlilikni o'zi ta'minlaydi, BGP — TCP hisobiga

## 3. Dinamik protokol qanday ishlaydi

Tarmoqdagi hamma router bir xil protokolni yuritishi kerak.



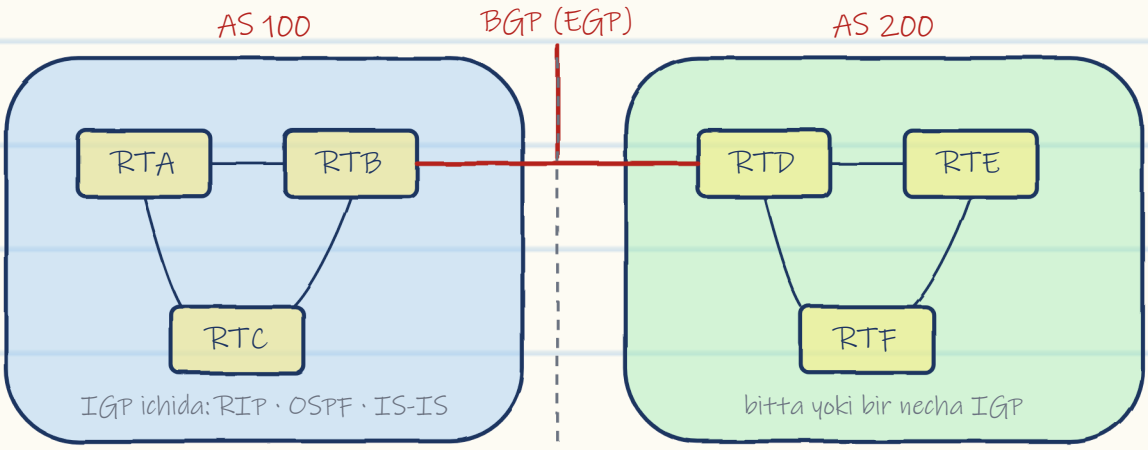

3-rasm. To'rt qadam: topish → almashish → hisoblash → kuzatish

- topish — o'zini bevosita ulangan segmentlarda tanitadi
- almashish — jadval qo'shnidan qo'shniga o'tadi
- hisoblash — algoritm next hop va metrikani beradi
- kuzatish — davriy paket bilan nosozlik aniqlanadi

Sabab: har protokolning o'z "tili" bor — turli protokol yuritgan routerlar bir-birini tushunmaydi.

## 4. IGP va EGP

Bo'linish ish doirasiga qarab: AS ichida — IGP, AS'lar orasida — EGP.



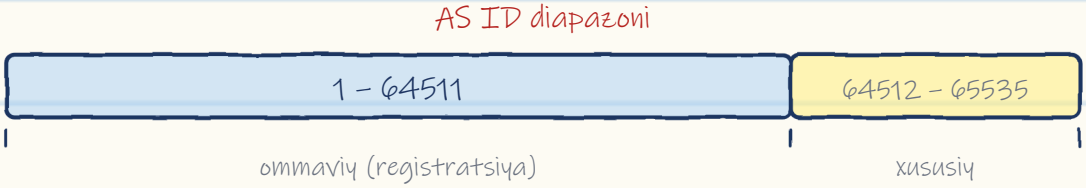

AS = umumiy siyosat, bitta boshqaruv hududi

tashqi dunyo AS'ni bitta butunlik deb ko'radi

AS raqamini IANA beradi — noyob

4-rasm. IGP — chegara ichida, EGP — chegaradan tashqariga

| | | |
|---|---|---|
| IGP | RIP · OSPF · IS-IS | AS ichida marshrut topadi |
| EGP | BGP | AS'lar orasida, siyosat va filtr |

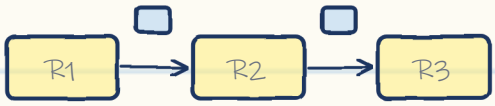



5-rasm. Ommaviy va xususiy AS raqamlari

- AS — umumiy marshrut siyosati, bitta boshqaruv hududi
- AS ichida bir necha xil IGP yurishi mumkin
- AS raqami noyob — IANA beradi

## 5. D-V va Link-State

**Distance-Vector**




qo'shniga butun jadvalni yuboradi

muhimi: masofa (hop) + yo'nalish

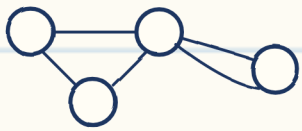
algoritm: Bellman-Ford

\+ oddiy, kam resurs

– sekin, halqa, hop 16

RIP · BGP

**Link-State**



faqat o'zgarishni (LSA) tarqatadi

LSDB → topologiya grafi

algoritm: Dijkstra (SPF)

\+ tez, halqasiz, masshtabli

– ko'p CPU va xotira

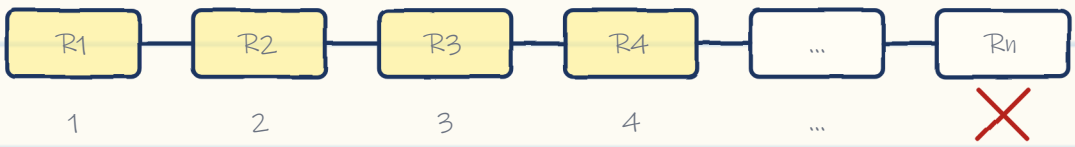
OSPF · IS-IS

6-rasm. Masofa va yo'nalish — yoki butun topologiya grafi

BGP ham D-V, lekin inkremental yangilash qiladi — polosa tejaladi.

| | D-V | Link-State |
|---|---|---|
| algoritm | Bellman-Ford | Dijkstra (SPF) |
| yangilash | RIP davriy · BGP inkremental | o'zgarishda, inkremental |

RIP: metrika = hop soni




hop chegarasi 16 — undan uzog'i yetib bo'lmaydi

shuning uchun RIP faqat kichik tarmoqda

7-rasm. D-V ning asosiy cheklovi — hop soni

## 6. Baholash ko‘rsatkichlari

| 5 ko‘rsatkich | |
|---|---|
| to‘g‘rilik | halqa bo‘lmasin — OSPF > RIP |
| konvergensiya | OSPF/BGP RIP’dan tezroq |
| narx | OSPF ko‘proq CPU va xotira |
| xavfsizlik | OSPF, RIPv2 — autentifikatsiya; RIPv1 yo‘q |
| masshtab | RIP kichik · OSPF katta · BGP butun dunyo |

8-rasm. Protokol tanlash — shu beshta ustundan qaraladi

## Xulosa

- Marshrutlash protokoli — hisoblaydi; marshrutlanuvchi (IP) — tashiydi
- RIP — UDP 520 · OSPF — IP 89 · BGP — TCP 179
- Ish tartibi: qo‘shnini topish → almashish → hisoblash → kuzatish
- IGP (RIP, OSPF, IS-IS) — AS ichida; EGP (BGP) — AS’lar orasida
- AS ID: 1–64511 ommaviy, 64512–65535 xususiy
- D-V — Bellman-Ford; Link-State — Dijkstra, LSA va LSDB

### Tekshirish

1. OSPF protokol raqami? — 89, IP ustida
2. RIP porti? — UDP 520
3. BGP porti va turi? — TCP 179, EGP va D-V
4. Xususiy AS raqamlari? — 64512–65535
5. Link-State algoritmi? — Dijkstra (SPF)